

# Course Project - Proposal

Robin Gibson, Andrej Scheuer, Lorenzo Celli, Oliver Saravanja

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| Name             | Student-ID | Mail                  |
|------------------|------------|-----------------------|
| Robin Gibson     | 20-119-467 | rgibson@ethz.ch       |
| Andrej Scheuer   | 20-917-829 | ascheuer@ethz.ch      |
| Lorenzo Celli    | 24-934-408 | lorenzo.celli@ethz.ch |
| Oliver Saravanja | 20-917-142 | osaravanja@ethz.ch    |

## 1 Feature list

### 1.1 Primary goals

These two features serve as the main focus of our project. We would be happy with the project if we implement these in a satisfactory way.

#### 1.1.1 FP8 Training w/ TransformerEngine

The trick will be to implement this in a way such that precision suffers minimally.

**Interest:** Oliver, Robin, Lorenzo

**Related work:** TransformerEngine quantization library

#### 1.1.2 Kernel Fusion & Liger-Kernels

By fusing various operations into the same kernel, we aim to improve the throughput of the model.

**Interest:** Robin, Andrej, Lorenzo

**Related work:** Liger-Kernel GitHub Repository

### 1.2 Stretch goals

If we are happy with our main goals, these features serve as additional work we can take on.

#### 1.2.1 Flash Attention v3

This should improve the throughput drastically, and it synergizes well with the FP8 quantization. As such, this makes sense for us to include in the project.

**Interest:** Robin, Andrej

**Related Work:** FlashAttention-3 Paper

#### 1.2.2 Leverage CPU Activation Offloading w/ TransformerEngine

As mentioned in the slides, the unique architecture of the GH200 lends itself very well to this feature, so in an effort to improve throughput drastically, we aim to implement this as part of our project. Due to doing FP8 training, somewhat familiar with TransformerEngine.

**Interest:** Oliver, Lorenzo

**Related work:** TransformerEngine CPU offload library